

Structure of solutions of Non-homogeneous linear ODEs, Continued

2.5 (Continued from worksheet 12). Linear homogeneous constant-coefficient ODEs with Order $n = 2$ and Characteristic Polynomial with complex conjugate pair of roots.

Case III: $\Delta < 0$.

Write

$$\lambda_{1,2} = -\frac{a_1}{2} \pm i\frac{\sqrt{-\Delta}}{2} = a \pm ib, \quad a = -\frac{a_1}{2}, \quad b = \frac{\sqrt{-\Delta}}{2} > 0.$$

The two complex solutions are

$$\hat{y}_{1,2}(t) = e^{(a \pm ib)t} = e^{at}(\cos(bt) \pm i \sin(bt)).$$

Both their real and imaginary parts are real solutions. Thus take

$$y_1(t) = \underline{\hspace{2cm}}, \quad y_2(t) = \underline{\hspace{2cm}}.$$

Is this an FSS? Compute the Wronskian at $t_0 = 0$.

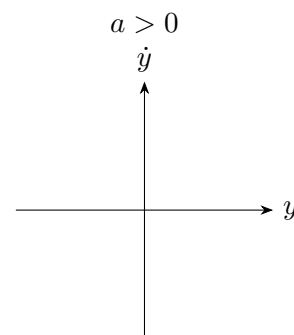
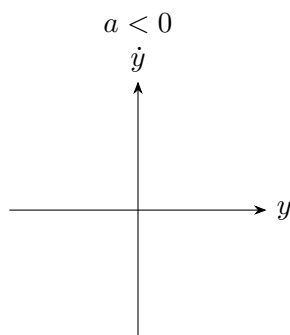
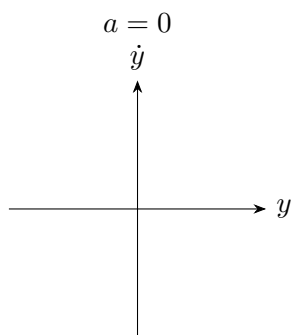
The real general solution is

$$y(t) = \underline{\hspace{4cm}}.$$

Classify the zero equilibrium using the real part a .

Condition	Stability of $\bar{x} = (0, 0)^T$
$a = 0$	
$a < 0$	
$a > 0$	

Sketch nearby orbits in the $(y, \dot{y})$ plane. Indicate direction.



3. Application: the free response of an RLC circuit

Example 13.1 – RLC circuit. Consider the capacitor-voltage equation

$$LC\ddot{v} + RC\dot{v} + v = v_s(t), \quad (1)$$

where $L > 0$, $C > 0$, and initially $R > 0$.

Rewrite the equation in terms of the capacitor charge $q = Cv$.

Since $v = q/C$, the charge satisfies

$$\underline{\hspace{10cm}} = v_s(t). \quad (2)$$

For the **free response**, set $v_s(t) \equiv 0$.

Find the characteristic equation, discriminant, and roots.

$$\underline{\hspace{10cm}} = 0, \quad \Delta = \underline{\hspace{10cm}}.$$
$$\lambda_{1,2} = \underline{\hspace{10cm}}.$$

Case I: $\Delta > 0$. Why are both roots negative?

Complete an FSS and the general solution:

$$\text{FSS} = \underline{\hspace{10cm}},$$
$$q(t) = \underline{\hspace{10cm}} \longrightarrow \underline{\hspace{10cm}} \quad (t \rightarrow \infty).$$

The circuit is called $\underline{\hspace{10cm}}$.

Case II: $\Delta = 0$. Complete the repeated-root solution.

$$\lambda_1 = \lambda_2 = \underline{\hspace{10cm}}, \quad \text{FSS} = \underline{\hspace{10cm}}.$$
$$q(t) = \underline{\hspace{10cm}}.$$

The circuit is called $\underline{\hspace{10cm}}$.

